

SUZUKI-A. LAB ORIENTATION – DAY 1

Welcome to Suzuki-A. Lab: Installing the Research OS

Research runs on design, not luck

Suzuki-A. Laboratory, Institute of Fluid Science, Tohoku University

UPDATE STARTING NOW

| Today's Agenda

- | | | |
|---|--------------------------------|--------|
| 1 | What is Research? | Part 1 |
| 2 | Research as Design | Part 2 |
| 3 | Paper Types & Responsibilities | Part 3 |
| 4 | The 5-Layer Structure | Part 4 |
| 5 | Grade-Specific Goals | Part 5 |
| 6 | Habits & Culture | Part 6 |

TODAY'S QUESTION

Is research Luck – or Design?

We value who you become through research more than the results

Lab Policy — we start with what matters most. Results → Human Growth. Research is a device for character building. We will come back to this at the very end of today.

| Lab Values – the four values of Suzuki-A. Lab

The criteria behind every activity. Everything today is an implementation of these four

1 Self-management

Ownership of your research is yours. Design your own time, plans, and health. The Research Log and Quarter Plan are tools of self-management. Don't wait to be managed.

2 Integrity

Be honest with the data, with people, and with yourself. Share unsuccessful results and stagnation openly. "Silence when stuck" is the opposite of integrity.

3 Respect

Respect for people, ideas, and time. In discussion, critique the Work — never attack the Peer. Come prepared, out of respect for others' time.

4 Collective Inquiry

Research is not a solo game. Paper Card and Inquiry Cycle form a common language that accelerates each other's inquiry. Culture is built by everyone.

Values are shown in daily actions, not on posters

Culture is not accidental. It is designed.

Research is Culture — Structure → Habit → Culture. The "Research OS" we install today is the blueprint of this culture.

| Is research Luck – or Design?

Passive research vs intentional research

Luck

- ⊗ Waiting for inspiration
- ⊗ Trial and error only
- ⊗ Results are random



Design

- ✓ Intentional inquiry
- ✓ Structured process
- ✓ Reproducible results

Blaming luck for your results ends today

| Same hours, different outcomes – why?

The Lost vs The Focused

The Lost

- Data without a claim
- Low contribution

The Focused

- They live and die by question design
- High impact

It's not about Talent. It's the OS.

| The "Data-First" trap: experiments first, meaning later

How research quietly goes off the rails

1

Collect data

Gather data first, without a question



2

Just run it

Exp / Sim — figure out the meaning
later



3

Get lost

No direction, no claim — wandering

Entering the forest without a map ends today

| The new standard: Suzuki-A. Lab OS

From talent-based to system-based



Systematization

Replace individual sense with a system anyone can run.



Common Language

Paper Card, Inquiry Cycle — one vocabulary for every lab member.



Reproducibility

Anyone can reproduce high-quality outcomes.

Aiming for overwhelming reproducibility

**A paper is not the end result.
It is a medium to receive "error".**

Goal of Day 1: shake your attitude toward research — accept the gap, and take on the responsibility of continuous updating

| Are you ready to update?

How this orientation is structured

DAY 1: Soul

- Philosophy
- Expectations
- Responsibility

DAY 2: Action

- Implementation
- Tools
- Practice

Mixing dilutes. Today, we start with the soul.

SECTION 01

What is Research?

Inquiry — not answer collection

01

**Inquiry is never "finished".
It keeps being updated.**

Research is Inquiry — not a project with a deadline, but a living, iterative cycle. There is no linear goal
(Research $\neq$ linear path)

| Research is not collecting answers – it refines questions

The Essence of Research

✕ Collecting answers

- ✕ Piling up answers — Answer Pile
- ✕ Aiming for a final answer



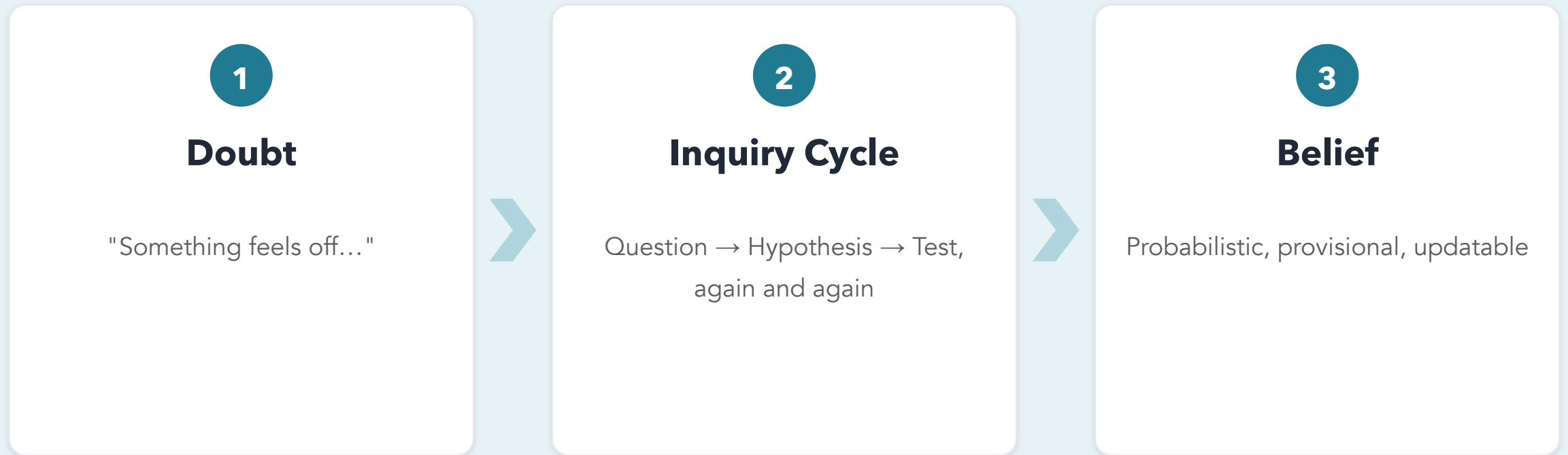
○ Refining questions

- ✓ Sharpening the question — **Refined Question**
- ✓ Toward a sharper, better-framed question

The goal is not a final answer – it is a sharper question

| Inquiry turns doubt into belief

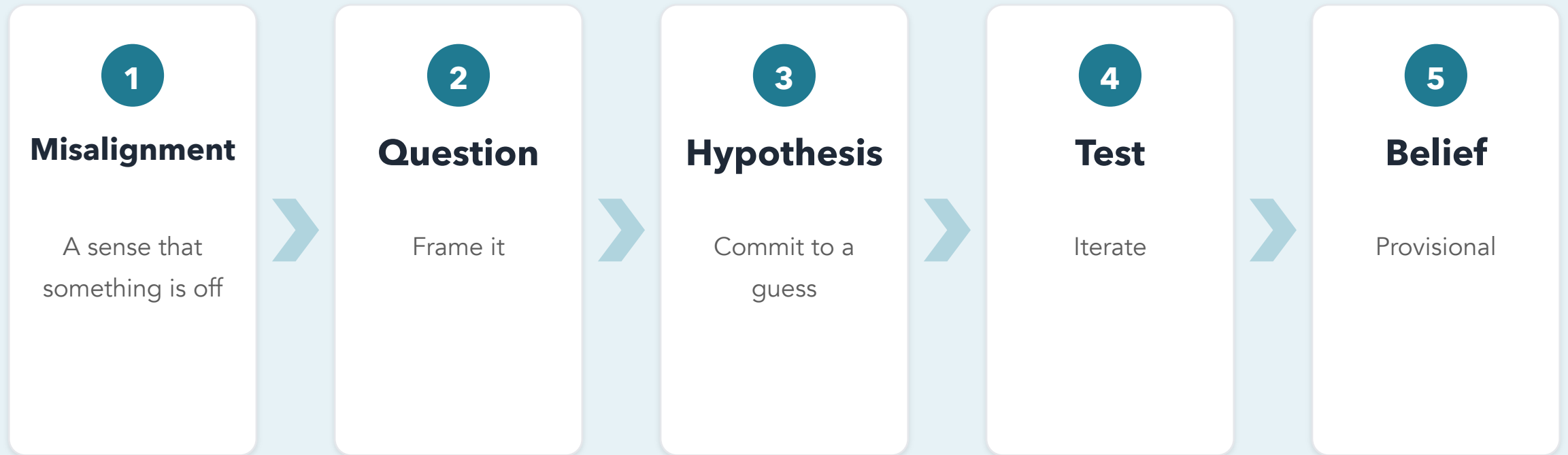
From Doubt to Belief



Keep the Inquiry Cycle turning ↻

| Inquiry Cycle: updating is the essence

"Theory $\neq$ Observation"



Provisional Belief – always open to revision

| Mismatch is the beginning – the gap generates inquiry

The gap is the starting point

Observation ≠ Understanding

- The gap between what we observe and what we understand generates questions

Ideal ≠ Reality

- The gap between the ideal and the real generates questions

GAP → Inquiry starts here

| Cognition shares the same structure

The brain works the same way



Cognition and Inquiry share one structure – Free Energy Principle (K. Friston)

The brain strives to minimize prediction error

Free Energy Principle (intuitive version) — large error → discomfort → action & update / small error → stable belief. Beyond the update lies a refined belief

| Belief is not "Truth". It is Probability.

What is a Belief?



Always provisional

The best estimate at this moment — nothing more.



Always updatable

New evidence rewrites it.



Strength varies with evidence

A probability distribution over hypothesis space.

Hold the most probable belief – and keep updating it

A paper is a medium that fixes questions and beliefs

Not a "final answer" — a snapshot of the best belief at this moment. Doubt → Inquiry → Paper (Fixed Belief)

| What happens when you fix a belief – why publishing matters

Consequences of Fixing

1

Paper

Fixing the belief — "This is our current best belief."



2

Critique

Peers can challenge, test, and verify



3

Next Question

On to the next inquiry

Each paper opens new inquiry cycles

Inquiry is submitting provisional belief for public update

Inquiry as Responsibility — research never ends; it keeps updating. "A paper is a medium to receive Feedback & Error."

SECTION 02

Research as Design

Research stops not because of lack of ability, but because of lack of order and structure

02

Research stalls not from lack of data, but from lack of direction

Research Stagnation — data increases, analysis proceeds, but what you want to show stays vague (no question)

| Failure pattern: Data-First

Start with data. Hope meaning emerges later.

1

Just experiment

 Move your hands first



2

Just analyze

 Analyze whatever came out



3

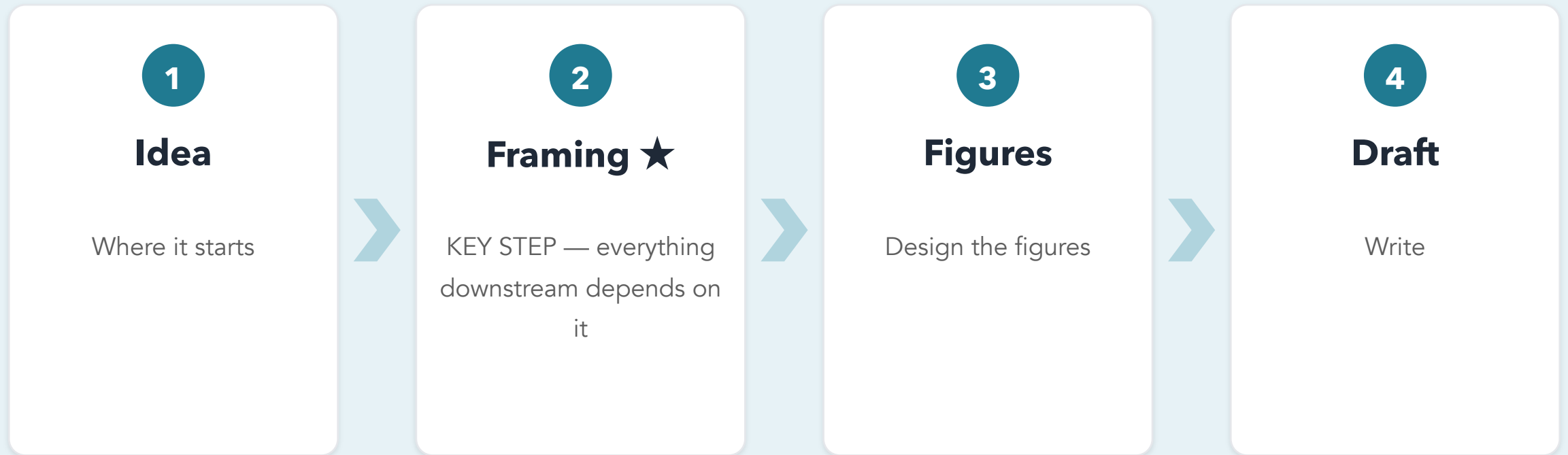
Meaning later

 Exploration without direction —
the wandering loop

This is not design. This is wandering.

| The correct order: Framing is the blueprint

Order is everything



Order is everything – Framing is the blueprint of your research

| What is Framing?

Only three things to write

- 1 **Core Question in one sentence**
- 2 **State the difference: "Compared to X, we Y."**
- 3 **Key contributions – three at most**

Don't write results. Don't write methods.

| Weak framing = total collapse

The same study, two questions

× Vague question

- × "We studied X and found something interesting..."
- × What's new? Under what conditions? Unclear.



○ Structured question

- ✓ "Compared to prior method X, our approach Y achieves **Z** under condition C."
- ✓ Difference, result, and condition — all in one sentence

State your question in the "Compared to X, we Y" form

| Every good research question fits a type

Question Archetypes

Archetype	Form of the question
Estimation	How much / many?
Mechanism	How / why does it work?
Validity	Does method X work?
Uncertainty	What are the limits?
Scale	Does it scale?
Dynamics	How does it change?
Design	How should we build X?

A question that cannot choose an archetype is vague

| Once the question is fixed, the required data is decided

Question determines everything — data follows structure

Question

Fixed first — structure calls in the data

Required Data

Back-calculated from the question

**Method /
Experiment**

Methods and experiments

Data comes later.

× Data → hope for meaning / ○ Question → required data

| 1 figure = 1 claim – figures are the claim itself

Figures are arguments, not illustrations

BAD ✕

- ✕ Claim A and Claim B crammed into one figure
- ✕ Readers can't tell what to take away



GOOD ○

- ✓ **One** claim per figure
- ✓ Claim B goes to the next figure

Read papers via figures first

| Structure defines productivity – not ability

It's architecture, not talent

High Structure

- Framing → Figures → Draft → Submit
- Design quality determines output

Low Structure

- Data → ??? → Repeat
- Same hours, no output

Same time investment, different output – back to why we need an OS

| Paper Card: the device that fixes structure

From uncertain idea (Doubt) to confident design (Belief)

1 Core Question

One sentence

2 Compared to X, we Y

State the difference from prior work

3 Key Contributions

Three at most

4 Main Figures (sketch)

Layout of the figures

5 Contribution Statement

Summary of the contribution

Structure before data – No Card → No Paper

SECTION 03

Paper Types & Responsibilities

Belief takes multiple forms

**Paper types are not ranks.
They differ only in scope of belief.**

Original / Review / Technical / Short — equal standing, no hierarchy. Same "belief", different zoom levels.

| Four basic formats

Four ways to fix a belief

1 Original

A new claim

2 Review

Reframing the questions

3 Technical

Methodological

4 Short

A limited claim

Formats are tools, not status

| Venues differ: one axis is speed, the other is rigor

Venues of Publication



Conference

Start a discussion. Speed first.



Journal

Rigorous verification. Rigor first.



Thesis

Integrated accountability.

Different venues, different roles – use them accordingly

A thesis is internal accountability.
A journal is external verification.

Why a journal is necessary — Inquiry completes in public space. Lab → Scientific Community

| The paper type follows the research phase

Research Phase and Paper Type



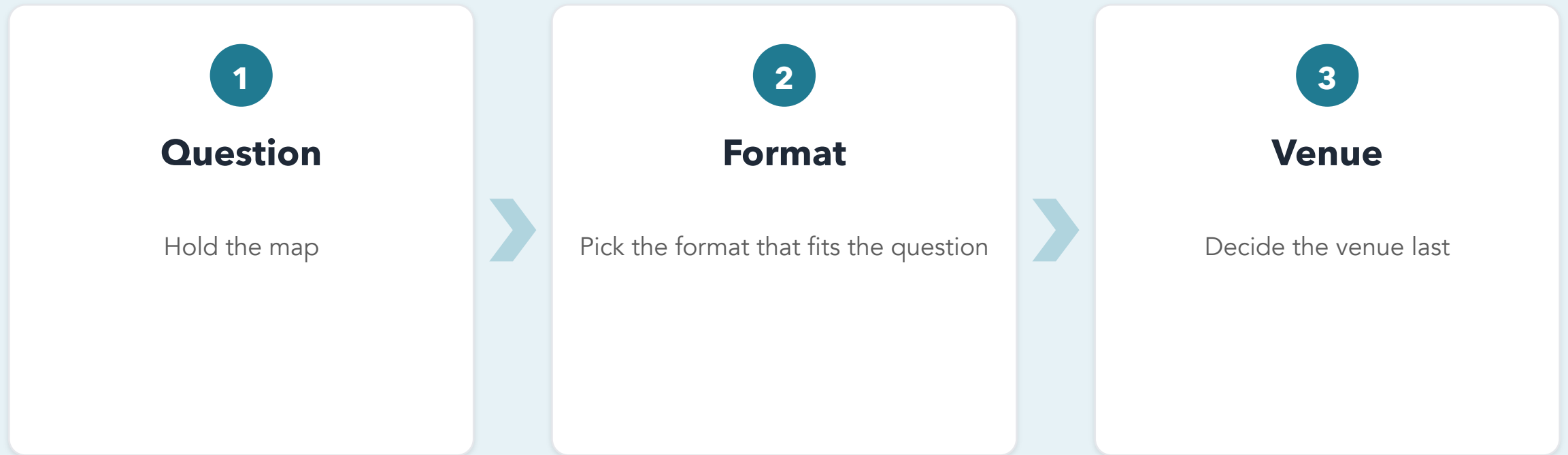
A PhD holds an axis – a Review is one form of it

**A thesis is not a page count.
It is a coherent belief system.**

Paper 1 → Paper 2 → Paper 3, connected into a Research Arc — details in the Quarter Plan

| Don't choose by format. Choose from your question.

The map-like order



Question → Format → Venue – reverse the order and you get lost

| Ethics is the qualification for submitting belief

What is Research Ethics? — Ethics as the Foundation of Belief



Data Integrity

Honest data and reproducibility.



Attribution

Credit and citation — show whose contribution it is.



Transparency

Disclose in a verifiable form.

A paper is an ethically submitted, verifiable belief – Claim, Ethics, Verify

SECTION 04

The 5-Layer Structure

Strategic organization of lab activities

04

| Lab activities are designed in five layers

The 5 Layers

1

Lab Forum

Value refinement space

2

Team Meeting

Tactical decision space

3

Student Interaction

Autonomy training space

4

Individual Meeting

Identity alignment space

5

Individual Work

The only place research moves

Each layer has its own role. Don't mix them.

| Four meetings, four roles

Layers 1 – 4



1. Lab Forum – where we polish the Why

Refine value and positioning. Constructive doubt and philosophical deep dives from diverse perspectives.



2. Team Meeting – decisions that move research forward

Paper Card → decision → next action



3. Student-only interaction – autonomy training

Technical consultation, verbalization training, pre-review



4. Individual Meeting – confirming your research axis

Growth direction and career design. **Not** for technical details or minor manuscript edits.

| The place for decisions, the place for sparring

Idea → Framing → Figures → Draft → Submission

Team Meeting = Decision

- Decides GO / NO-GO
- Move to the next phase, or go back

Lab Forum = Sparring

- Break and polish questions, positions, expressions
- Not a place for conclusions

Decisions move the phases; sparring polishes them

**Research doesn't move in meetings.
It moves at your desk.**

5. Individual Work — required habits: daily Research Log / weekly Paper Card / monthly Core Question. Meeting is for alignment. Work is for progress.

| Core Principle – four places, four purposes

Always know what each place is for

1 Team Meeting

Tactics — Tactical Decision

2 Lab Forum

Value — Value Refinement

3 Individual Meeting

Axis — Identity Alignment

4 Individual Work

Progress

Never mix the purposes – that is how the five layers run

SECTION 05

Grade-Specific Goals

Academic responsibility & grade-specific goals

05

More responsibility brings more freedom

Dual-axis model — horizontal: research responsibility / vertical: community responsibility. B4 → Master
→ PhD moves toward the upper right

| Responsibilities by grade

B4 / Master / PhD



B4: Experience & Honesty

Run one full research cycle. Share results and struggles honestly.
Completion over perfection.



Master: Completion & Mentorship

Submit one Original article. Support and guide the B4s. The responsibility to finish the race.



PhD: Definition & Culture

Define a unique research axis.
Protect and evolve the lab culture. A Review is one way to establish the axis.

Grow along both axes – research and community

| Expectations are explicit

Clear Expectations — This is NOT optional. This is the baseline.

B4

1 full cycle

Experience the whole research cycle
once

Master

1 Original

Submit one first-author paper

PhD

Axis + Culture

Define the axis, carry the culture

Baseline, not stretch goals

| Quarter Plan: not a progress report – structural visualization

Structural Reflection — design in 90-day units

What it is

- Strategy designed in 90-day quarters
- The one-year Research Arc split into Q1–Q4
- A visualization of your thinking structure

Why 90 days

- Research is a marathon; design in sprints
- For B4, the goal is one full cycle, not perfection
- Short spans expose structural drift early

Strategy is quarterly – run the marathon on sprint-sized designs

Locate yourself before moving

Identity Snapshot — YOU ARE HERE. Forward progress cannot be designed from a vague location.

| The B4 year: back-calculate from the deadlines

Not a year of research only — external events squeeze research

Q1 Apr-Jun

Fix the theme — in parallel with coursework

Q2 Jul-Sep

Framing — entrance exams and job hunting squeeze research

Q3 Oct-Dec

Figures — finalize the figures

Q4 Jan-Feb

Thesis wrap-up

Submit the thesis — one full Research Arc completed

APR → FEB: design backwards from the external events

| Typical B4 failure patterns

B4 Failure Patterns

- 1 Stalled by graduate entrance exams**
- 2 Silence when stuck**
- 3 Waiting for "perfection" to share**
- 4 Total collapse from multi-tasking**

The Quarter Plan exists to prevent exactly this

| Evaluation is a by-product of structural thinking

Evaluation Criteria — what is evaluated

- 1 Understanding**
- 2 Logical presentation**
- 3 Q&A performance**
- 4 Document structure**

Complete the Arc → skills will follow

| The Master's two years are an extended Arc

The race is decided in M1

M1: Exploration & Design

- Exploration and strategy
- Focus on Framing
- **Axis fixed** by mid-M1

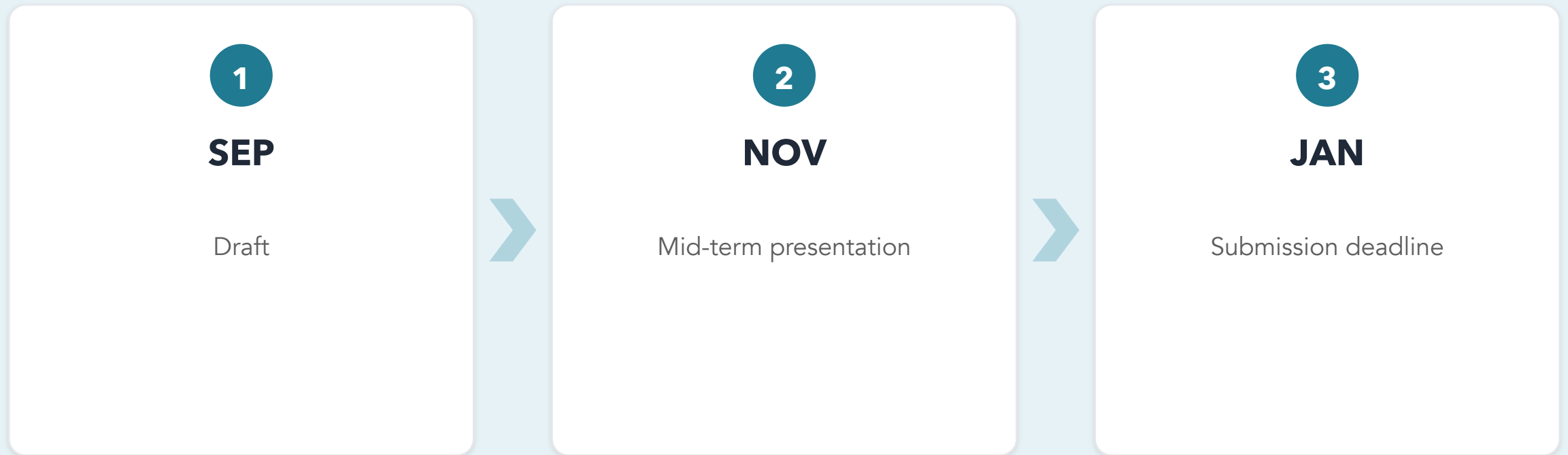
M2: Convergence & Completion

- The acceleration zone
- Those who panic in M2 didn't design in M1

Focus on Framing in M1. Fix your Axis.

| The reality of M2: time disappears suddenly

Reality of M2



Time disappears suddenly – that is why you design in M1

| The PhD's three years: build, expand, integrate

PhD: Axis Construction (3 Years)

D1: Build the axis

Axis construction — Review paper (externalize the hypothesis)

D2: Expand

Original expansion — deepen methods, gain independence

D3: Integrate

Integration & positioning

Structural design of the thesis

"Axis-less PhDs will drift."



SECTION 06

Essential Habits & Culture

Research moves through daily practice

06

The difference is not talent – it is the density of daily routine

Essential Habits — you are the one who accumulates the days

1

Research Log & Thinking Log

Every day

2

Paper Card (Paper Board)

Every week — keep updating the claim

3

Team Meeting / Lab Forum

Regular — decisions and sparring

4

Individual Meeting

Periodic — axis check

5

Quarter Plan

Every 90 days — structural visualization

Research moves through daily practice

| Research Log and Thinking Log

What is the Research & Thinking Log?

Research Log

- **What you did** today
- Facts

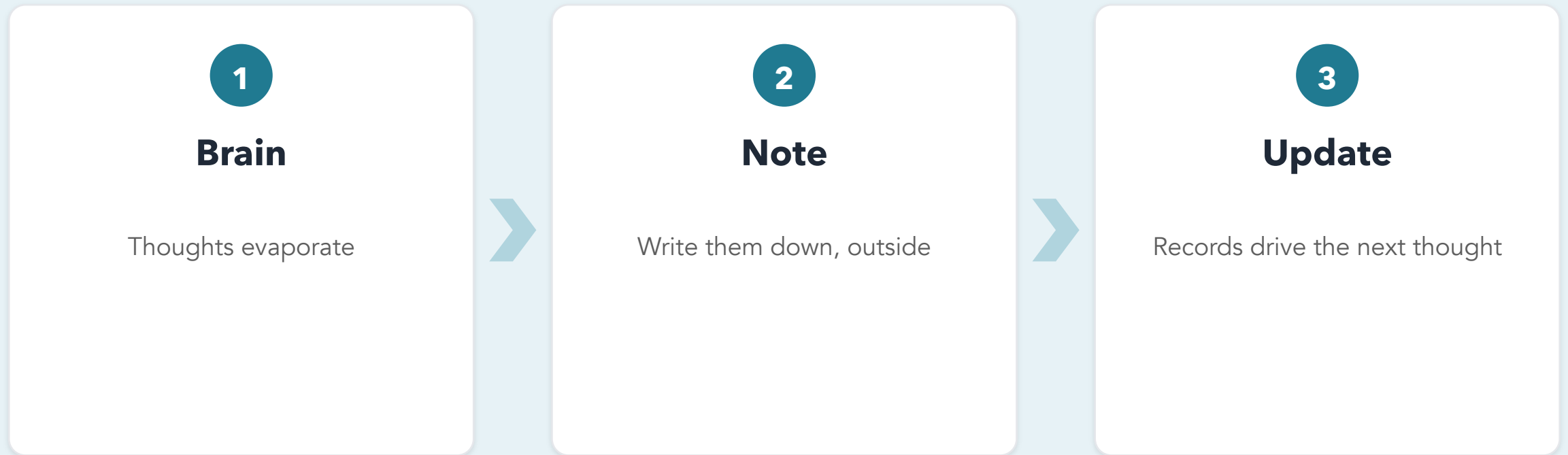
Thinking Log

- **What you thought** today
- Interpretation

Research progresses through: Fact → Interpretation

| Thinking happens outside the brain

Why log? — research runs on external records, not memory



Those who do not write repeat the same mistakes

| How to log

Log → Paper Arc

- 1 One entry every day**
- 2 No need for perfection**
- 3 Write hypotheses, facts & interpretation**

Writing drives the research forward – demo next week

We value who you become through research more than the results

Lab Policy (reprise) — back to the promise we made at the start. Results → Human Growth. Research is a device for character building.

| Lab Values – once more, what matters most

Your standard of action from tomorrow. Everything today was an implementation of these four

1 Self-management

Ownership of your research is yours. Design your own time, plans, and health. The Research Log and Quarter Plan are tools of self-management. Don't wait to be managed.

2 Integrity

Be honest with the data, with people, and with yourself. Share unsuccessful results and stagnation openly. "Silence when stuck" is the opposite of integrity.

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Research is not a solo game. Paper Card and Inquiry Cycle form a common language that accelerates each other's inquiry. Culture is built by everyone.

Values are shown in daily actions, not on posters

Culture is not accidental. It is designed.

Research is Culture (reprise) — Structure → Habit → Culture. Everything we discussed today is a component for designing this culture.

Are you ready to update?

The Research OS is installed — now keep updating, every day

Questions anytime: Lab Forum / Team Meeting / Individual Meeting